Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence
- Understanding Linear Independence in Vector Spaces — Reading • 18 min
- Linear Independence, Basis, and Rank — Ungraded App Item • 20 min
- Linear Independence — Reading • 30 min

Linear Mapping
- Exploring Transformations: Understanding Linear Mappings — Reading • 1h
- Linear Mapping: Part 1 — Video • 8 min
- Linear Mapping: Part 2 — Video • 8 min
- Linear Mapping — Reading • 5 min
- Matrix Representation — Reading • 6 min
- Practice Quiz: Linear Mapping — Graded Assignment • Grade: 100%

Lesson 3: Eigenvalues and Eigenvectors
- Introduction to Eigenvalues and Eigenvectors — Reading • 40 min
- Eigenvectors and Eigenvalues: Examples and Applications — Reading • 30 min

Lesson 4: Cholesky Decomposition
- Cholesky Decomposition — Reading • 5 min
- **Solved Examples** — Reading • 10 min

Lesson 5: Singular Value Decomposition
- Introduction to Singular Value Decomposition (SVD) — Reading • 28 min
- Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors — Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Solved Examples

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Consider the symmetric positive definite matrix:

$$A = \begin{bmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{bmatrix}$$

We want to find its Cholesky decomposition $A = LL^T$, where L is a lower triangular matrix.

1. **Define the Structure of** L: Since L is a lower triangular matrix, we can write:

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

2. **Calculate** l_{11}: From $A_{11} = l_{11}^2$:

$$l_{11} = \sqrt{A_{11}} = \sqrt{4} = 2$$

So, $l_{11} = 2$.

3. **Calculate** l_{31}: From $A_{31} = l_{21} \cdot l_{11}$:

$$l_{21} = \frac{A_{21}}{l_{11}} = \frac{12}{2} = 6$$

So, $l_{21} = 6$.

4. **Calculate** l_{31}: From $A_{31} = l_{31} \cdot l_{11}$:

$$l_{31} = \frac{A_{31}}{l_{11}} = \frac{-16}{2} = -8$$

So, $l_{31} = 8$.

5. **Calculate** l_{22}: From $A_{22} = l_{21}^2 + l_{22}^2$:

$$l_{22} = \sqrt{A_{22} - l_{21}^2} = \sqrt{37 - 6^2} = \sqrt{37 - 36} = 1$$

So, $l_{22} = 1$.

6. **Calculate** l_{32}: From $A_{32} = l_{31} \cdot l_{21} + l_{32} \cdot l_{22}$:

$$l_{32} = \frac{A_{32} - (l_{31} \cdot l_{21})}{l_{22}} = \frac{-43 - (-8 \cdot 6)}{1} = \frac{-43 + 48}{1} = 5$$

So, $l_{32} = 5$.

7. **Calculate** l_{33}: From $A_{33} = l_{31}^2 + l_{32}^2 + l_{33}^2$:

$$l_{33} = \sqrt{A_{33} - (l_{31}^2 + l_{32}^2)} = \sqrt{98 - ((-8)^2 + 5^2)} = \sqrt{98 - (64 + 25)} = \sqrt{9} = 3$$

So, $l_{33} = 3$.

Result:

Now, we have:

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 1 & 0 \\ -8 & 5 & 3 \end{bmatrix}$$

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